MDQ Specification Audit: multiple-selection: mdq-spec

Date: 2026-10-15

Scope: The `multiple-selection` question type: `docs/question-types/multiple-selection.md`, `schema/multiple-selection.yaml`, its grammar, and the shared specs it depends on (`docs/references/grammar.md`, `docs/references/patterns.md`, `docs/question-types/question-base.md`, `docs/exam.md`, `docs/responses.md`).

Methodology

`scripts/spec_audit.py` passed all 14 steps before the audit (schema bundle, docs, corpus, Additional Rules tables, lint codes, frontmatter tables, schema agreement of mdq-py).

The lead auditor read the shared specs and `docs/question-types/multiple-selection.md` and recorded contradictions, ambiguities, omissions and inconsistencies.

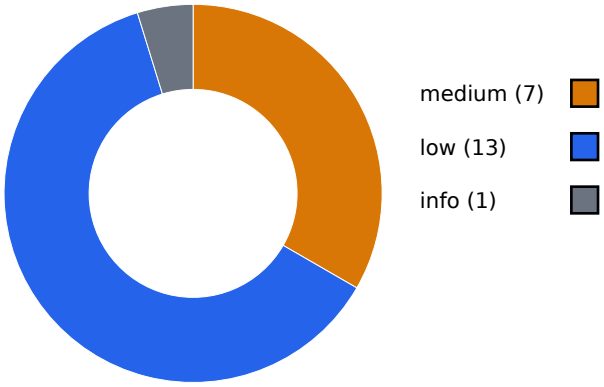
One subagent turned the grammars of `docs/references/grammar.md` and of multiple selection into Lark grammars and tested them with accepted and rejected inputs derived from the stated rules.

An independent subagent reviewed the multiple-selection document against `schema/multiple-selection.yaml`, ran each complete example through `mdq validate`, and checked the Additional Rules table against the MUST, SHOULD and MAY statements of the text.

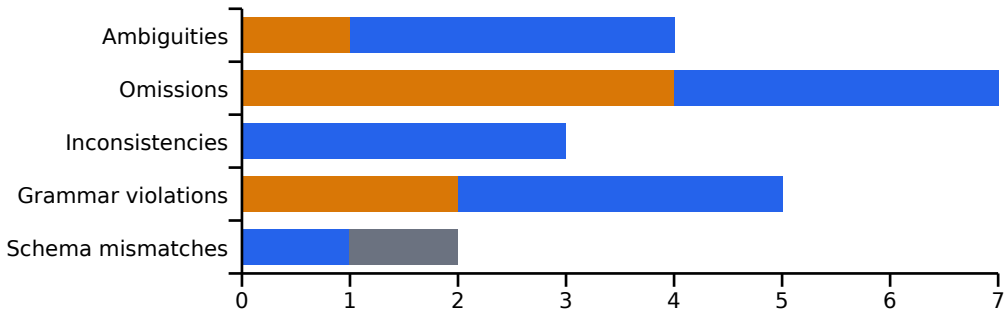
Executive summary

Severity	#
CRITICAL	0
HIGH	0
MEDIUM	7
LOW	13
INFO	1
Total findings	**21**

Findings by severity



Findings by category



Strengths and weaknesses

Strengths

✓ **Contradictions:** The grading table of multiple selection agrees with its formulas in every row, with exam.md#penalty (symmetric = 2 * partial - 1) and with the grading corpus. (docs/question-types/multiple-selection.md:125-135, docs/exam.md:314-321, examples/grading/multiple-selection.yaml)

✓ **Omissions:** Responses state the unticked-means-false rule, the equivalence of an absent key and `false`, and the difference between `null` (skipped) and `{}` (answered none), consistently with the Feedback and Grading sections. (docs/responses.md:49-72, docs/responses.md:192-210, docs/question-types/multiple-selection.md:76-92)

✓ **Inconsistencies:** The `grading` and `shuffle` inheritance of multiple selection uses the same wording, defaults and exam-map key as multiple choice and exam.md. (docs/question-types/multiple-selection.md:30-34, docs/exam.md:278-300, docs/exam.md:327-352)

✓ **Grammar violations:** The multiple-selection `value` rule matches the stated markers exactly. Tested with Lark: it accepts `[ ]`, `[  ]`, `[\t]`, `[x]`, `[X]`, `[ x ]`, `[ X ]` and `[\tx\t]` (every form in examples/valid/multiple-selection/uppercase-x.mdq.md), and rejects `[*]`, `[?]`, `[50%]`, `[v]`, `[y]`, `[xx]`, `[ x x ]`, `[U+00A0]` and `[U+00A0x]`, which agrees with the `foreign-choice-marker` examples (examples/invalid/multiple-selection-forced-star.mdq.md, multiple-selection-unknown-marker.mdq.md) and with the rule that U+00A0 is text, not `ws`. (docs/question-types/multiple-selection.md:41-45)

✓ **Grammar violations:** The CHOICE_ID token and its position in the item rule behave as the prose states. Tested with Lark: `[x] [amazonas] Amazonas` gives id `amazonas`; `[x] [brasilia]` gives id `brasilia` and empty text (the documented `blank-text` case); `[x] [IBGE](https://ibge.gov.br) site`, `[x] [amazonas][1] text` and the collapsed link `[x] [brasilia][]` give no id; `[x] [a b] text` and `[x] [-a] text` give no id because the content is not a SLUG. (docs/references/grammar.md:62-86)

✓ **Grammar violations:** With a stub `lazy_line` that excludes marker lines (see F10), the shared `observations` rules accept and reject what the prose states. Tested with Lark as the tail of the multiple-selection item: feedback only, comment only, feedback then comment, comment then feedback, a blank line between the text and the first block and between the two blocks, the grammar.md example with a lazy line and an empty > marker line, a `![img](x.png)` line as a lazy line of the feedback, and `! [IBGE](...)` as a comment are accepted. Two feedback blocks separated by a blank line, feedback-comment-feedback, two comment blocks, and a paragraph after the feedback are rejected. (docs/references/grammar.md:98-137)

✓ **Grammar violations:** The `nl` and `ws` rules hold for the multiple-selection item. Tested with Lark: items with LF, CRLF and CR line endings parse identically, and `ws` does not match U+00A0. A last line without a line ending is rejected by the grammar, which the prose covers by reading it as if it ended with `nl`. (docs/references/grammar.md:27-38)

✓ **Schema mismatches:** The schema Choice requires `text` and `correct`, matching the REQUIRED `correct` in the spec, and its feedback description matches the wrongly-judged rule of the Feedback section. (schema/multiple-selection.yaml:77-100)

✓ **Inconsistencies:** The grading table values match examples/grading/multiple-selection.yaml for all three strategies, and exam.md Penalty quotes the same values. (docs/question-types/multiple-selection.md:125-132)

✓ **Omissions:** The corpus covers uppercase and spaced x values, an all-blank list inferred without `type`, forced `[*]` and unknown `[?]` foreign markers, and every schema rule of the table. (examples/valid/multiple-selection/uppercase-x.mdq.md:8-12)

✓ **Omissions:** The complete example document of the spec validates with mdq-py without errors or warnings. (docs/question-types/multiple-selection.md:9-17)

Central risks

• The choice-item grammars leave to prose the rules that decide whether a line is a choice, a feedback line or text (indentation limits, lazy lines, the generic value bracket). mdq-py follows the looser grammar in several of these cases, so the prose is the only defense against drift between implementations.

• multiple-selection.md imports multiple-choice.md#choices by reference. An edit to the multiple-choice text can change the rules of multiple selection without anyone reviewing that effect.

Detailed findings

Contradictions

No findings in this category.

Ambiguities

Severity	Location	Description
MEDIUM	docs/question-types/multiple-selection.md:60-61	**F3** Choices section does not say which parts of multiple-choice#choices apply
LOW	docs/question-types/multiple-selection.md:120-132	**F5** The T, F and _ notation of the grading table is not defined
LOW	docs/question-types/multiple-selection.md:137-141	**F6** 'no way to explicitly leave it unmarked' reads as the opposite of its intent
LOW	docs/question-types/multiple-selection.md:100-102	**F7** 'When neither ... and outside an exam' joins two cases with 'and'

F3: Choices section does not say which parts of multiple-choice#choices apply

Location: docs/question-types/multiple-selection.md:60-61

```
Choices follow the same structure as [Multiple choice questions](multiple-choice.md#choices),
respecting the rules for the allowed `values` inside the initial square brackets.
```

Description: The referenced section mixes shared rules (text, id, feedback, comment, AST mapping table, duplicate text, visual equivalence, derived ids) with rules that are only true for multiple choice: the `score` row of the AST table, 'A choice is correct when its score is 1', 'the student still picks a single choice', the score scale and the `many-correct-choices` lint. The sentence 'respecting the rules for the allowed values' excludes only the value rule. A reader must guess that `score` is replaced by `correct` and that the single-pick sentence does not apply.

Impact: Readers and porters can apply multiple-choice-only rules (score derivation, single pick, many-correct-choices) to multiple selection.

Suggested fix: List the shared rules explicitly (text, id, feedback, comment, duplicates, visual equivalence, derived ids) and say that `correct` replaces the `score` row of the AST table; or move the shared choice rules to one shared section that both types reference.

F5: The T, F and _ notation of the grading table is not defined

Location: docs/question-types/multiple-selection.md:120-132

```
| `[T, F, _, _]` | `[T, F, F, F]` |        3         |   0.75    |       0.0        |     0.5
    |
```

Description: The table writes markings with T, F and _, and expands _ to F. The document says (line 140) that a student has no way to leave a choice unmarked, so a reader cannot tell what F is as opposed to _. The distinction only makes sense with responses.md, where F is an explicit `false` key and _ is an absent key. The document does not say so.

Impact: Readers can think that multiple selection has a third state, which the same document denies.

Suggested fix: Either write the markings as ticked sets ({a, c}, {a}, {}) and drop the 'Expands to' column, or say before the table that T and F are the `true` and `false` values of the response mapping and _ is an absent key (see responses.md#multiple-selection). Recommended: ticked sets, since both forms are the same response.

F6: 'no way to explicitly leave it unmarked' reads as the opposite of its intent

Location: docs/question-types/multiple-selection.md:137-141

Mechanical fix

```
There
is no way to explicitly leave it unmarked. If that is necessary, prefer using
[true/false](true-false.md) questions.
```

Description: Leaving a choice unmarked is exactly what an unticked checkbox does. The intended meaning is that a student cannot abstain on a choice: every choice gets a verdict. 'it' also has no clear referent (the choice? the question?).

Impact: Minor confusion about the meaning of an unticked choice.

Suggested fix: Reword: 'A student cannot abstain on a single choice: an unticked choice is a false verdict. If abstention is needed, use true/false questions.'

F7: 'When neither ... and outside an exam' joins two cases with 'and'

Location: docs/question-types/multiple-selection.md:100-102

Mechanical fix

```
When neither
the question nor the exam sets a strategy, and outside an exam, the strategy
is `"symmetric"`.
```

Description: Read literally, the default applies only when both conditions hold: no strategy is set and the question is outside an exam. These are separate cases: an exam that sets nothing, and no exam. The sentence also does not cover an exam whose `grading` is a map without a `multiple-selection` key, which exam.md resolves to `symmetric`. The same sentence is in multiple-choice.md:229-231.

Impact: Minor; the footnote settles it.

Suggested fix: Write: 'The strategy is `"symmetric"` outside an exam, and in an exam that does not set one for multiple selection (no `grading`, or a map without a `multiple-selection` key).' Apply the same wording in multiple-choice.md.

Omissions

Severity	Location	Description
MEDIUM	docs/question-types/question-base.md:213-216	**F1** Type inference has no rule for a list of blank and unknown values

Severity	Location	Description
MEDIUM	docs/question-types/multiple-selection.md:42-44	**F2** Empty brackets `[]` in a choice list have no rule
MEDIUM	docs/question-types/multiple-selection.md:5-7	**F14** Multiple selection does not state how a choice list is inferred as multiple selection
MEDIUM	docs/question-types/multiple-selection.md:156	**F25** No multiple-selection example has a second feedback or comment block, and mdq-py merges them
LOW	docs/question-types/multiple-selection.md:158-159	**F4** Text never states the no-correct-choice and all-choices-correct lints
LOW	docs/question-types/multiple-selection.md:48-50	**F26** Markdown-level blank-text and text-normalization cases are only covered by YAML examples
LOW	examples/grading/multiple-selection.yaml:17-115	**F27** Grading corpus has no question with zero or all correct choices, or with two choices

F1: Type inference has no rule for a list of blank and unknown values

Location: docs/question-types/question-base.md:213-216

```
1. a value is `*` or a percentage: multiple choice;
2. a value is `x` or `X`: multiple selection;
3. a value is a letter: true/false;
4. every value is blank: multiple selection.
```

Description: A list such as `* [ ] Manaus` / `* [?] Santos` matches none of the four rules: no value is *, a percentage, x or a letter, and not every value is blank. The spec does not say which type the list takes, so it does not say against which value rule `[?]` is checked. mdq-py reads it as multiple selection and reports `foreign-choice-marker` on `[?]`. The corpus only covers `[?]` next to an `[x]` (`examples/invalid/multiple-selection-unknown-marker.mdq.md`), where rule 2 applies.

Impact: Two implementations can infer different types, or none, and report different codes for the same document.

Suggested fix: Relax rule 4 to 'no other rule applies: multiple selection' (what mdq-py does), and add an invalid example with only blank and unknown values.

Exploitability conditions: A choice list without `type`, whose only values are blank and unknown markers.

F2: Empty brackets `[]` in a choice list have no rule

Location: docs/question-types/multiple-selection.md:42-44

```
value : ws
      | ws? ("x" | "X") ws?
```

Description: `value` needs at least one `ws`, so `* []` Santos has no value. The spec does not say what such an item is: a foreign marker (`foreign-choice-marker`), an item that makes the list not a choice list (then a forbidden list in the stem, `forbidden-block-element`), or a `parse-error`. mdq-py and the corpus (`examples/invalid/multiple-choice-empty-brackets.mdq.md`) use `parse-error`, which no text of the spec states. There is no multiple-selection example.

Impact: `[]` is a likely typo for `[ ]`. Implementations can disagree on the code, and one could read the list as prose and drop the question body.

Suggested fix: State in question-base.md, Type inference, that `[]` in a choice list is a `parse-error`, as mdq-py does, and add a multiple-selection invalid example.

F14: Multiple selection does not state how a choice list is inferred as multiple selection

Location: docs/question-types/multiple-selection.md:5-7

```
Multiple selection questions are represented by an unordered list with an ASCII-art
rendition of a checkbox. The correct answers are marked with `[x]` and incorrect
ones are left blank.
```

Description: multiple-choice.md:6-9 says when a list is inferred as multiple choice and links to type inference. multiple-selection.md says nothing: it does not state that a list with an x/X value, or a list whose values are all blank, is inferred as multiple selection (question-base.md:214-216), nor that a * or percentage value makes the list multiple choice instead. The all-blank case is used by examples/valid/multiple-selection/none-correct.mdq.md, which has no `type`.

Impact: A reader of this page cannot tell that a list with no `[x]` is still multiple selection, or that one `[*]` turns the whole list into multiple choice.

Suggested fix: Add a sentence after the example that summarizes the inference rules for multiple selection and links to question-base.md#type-inference, as multiple-choice.md does.

F25: No multiple-selection example has a second feedback or comment block, and mdq-py merges them

Location: docs/question-types/multiple-selection.md:156

```
| choices[]          | error    | parse-error               | carries at most one feedback
block and one comment block  |
```

Description: grammar.md (One block of each kind) says a second block of the same kind is a `parse-error`, also when a blank line or a block of the other kind separates them. No example under examples/invalid/ exercises this for a multiple-selection choice (only ordering-two-feedback-blocks exists). mdq-py accepts both `> one / blank line / > three` and `> one / ! c / > three` under one choice and joins the feedback into 'one three', with no error.

Impact: The rule is not tested, and the reference implementation violates it without the suite noticing.

Suggested fix: Add invalid examples for a choice with two feedback blocks (separated by a blank line, and by a comment block), expecting `parse-error`.

Exploitability conditions: mdq-py `mdq validate` on a choice with two > blocks.

F4: Text never states the no-correct-choice and all-choices-correct lints

Location: docs/question-types/multiple-selection.md:158-159

```
| choices[].correct  | info     | no-correct-choice          | at least one choice should be
correct                      |
| choices[].correct  | info     | all-choices-correct        | not every choice should be
correct                         |
```

Description: The Additional Rules table lists both info rules, but no sentence of multiple-selection.md states them with a MAY or explains why they matter. The text in multiple-choice.md about these codes refers to scores and to a single pick, so it does not carry over. For multiple selection the reason differs: with no correct choice, the empty response {} scores 1; with every choice correct, ticking all scores 1.

Impact: The rule exists only in the table; the level and the rationale cannot be checked against the text.

Suggested fix: Add a paragraph to Choices: any number of choices MAY be correct, including none or all; implementations MAY report `no-correct-choice` and `all-choices-correct` at the info level, with the reason above.

F26: Markdown-level blank-text and text-normalization cases are only covered by YAML examples

Location: docs/question-types/multiple-selection.md:48-50

```
In
`* [x] [brasilia]`, `brasilia` is the id and the text is empty, which is a
`blank-text` error.
```

Description: The corpus has blank-text and duplicate-choice-text examples for multiple selection only in YAML (examples/invalid/model-only/multiple-selection-*.yaml). No `.mdq.md` example covers the `* [x] [brasilia]` case the spec gives, a choice with only feedback markers, two texts that differ only in whitespace, a multi-line (lazy line) choice text, or multi-line feedback. multiple-choice has `.mdq.md` examples for the blank-text cases.

Impact: The parser path that produces these values for multiple selection is not tested.

Suggested fix: Add `.mdq.md` invalid examples for `* [x] [id]` with no text and for whitespace-only duplicate texts, and a valid example with multi-line choice text and feedback.

F27: Grading corpus has no question with zero or all correct choices, or with two choices

Location: examples/grading/multiple-selection.yaml:17-115

```
questions:
  # The table in multiple-selection.md, "Grading": answer key [T, F, T, F].
```

Description: All grading cases have at least one correct and one incorrect choice. No case covers an answer key with no correct choice (where `ticks: []` scores 1 under every strategy) or with every choice correct, both of which the spec allows (rules at multiple-selection.md:158-159 are only info), nor the two-choice minimum.

Impact: Edge cases of the grading formulas that the spec allows are not checked against implementations.

Suggested fix: Add grading cases with keys `{a: false, b: false}` and `{a: true, b: true}` for the three strategies.

Inconsistencies

Severity	Location	Description
LOW	docs/question-types/multiple-selection.md:39	**F8** Body does not state the two-choice minimum, unlike multiple choice
LOW	docs/question-types/multiple-selection.md:157-160	**F21** Rules table uses 'should' for info-level rules
LOW	docs/question-types/multiple-selection.md:158-159	**F22** Rules table field for count rules is a choice field, but the code reports on the list

F8: Body does not state the two-choice minimum, unlike multiple choice

Location: docs/question-types/multiple-selection.md:39

Mechanical fix

```
Body consists of an unordered list of items. Each item follows the grammar:
```

Description: multiple-choice.md:41 says 'an unordered list of at least two items'. multiple-selection.md states the minimum only in the Additional Rules table (schema `minItems: 2`).

Impact: Minor inconsistency between parallel documents.

Suggested fix: Write 'an unordered list of at least two items'.

F21: Rules table uses 'should' for info-level rules

Location: docs/question-types/multiple-selection.md:157-160

```
| choices[].text     | info     | visually-identical-choices | should not be visually
equivalent to another choice[^2]   |
| choices[].correct  | info     | no-correct-choice          | at least one choice should be
correct                      |
| choices[].correct  | info     | all-choices-correct        | not every choice should be
correct                         |
| choices[].id       | info     | missing-choice-id          | should be defined in the
document, instead of derived[^3] |
```

Description: By the spec convention, SHOULD maps to `warning` and MAY to `info`. The source prose (multiple-choice.md:118, 140, 164) says implementations MAY report these cases, so the level is info, but the Rule column phrases them as 'should'. The same phrasing is in multiple-choice.md:288-292.

Impact: A reader that maps the wording to a level gets `warning`, not `info`.

Suggested fix: Phrase the info rules as observations, for example 'is visually equivalent to another choice', 'no choice is correct', 'every choice is correct', 'has no id of its own'.

F22: Rules table field for count rules is a choice field, but the code reports on the list

Location: docs/question-types/multiple-selection.md:158-159

Mechanical fix

```
| choices[].correct  | info     | no-correct-choice          | at least one choice should be
correct                      |
```

Description: `no-correct-choice` and `all-choices-correct` are properties of the whole list. The Field column gives `choices[].correct`, but the corpus reports them at path `["choices"]` (examples/valid/multiple-selection/none-correct.lint.json, all-correct.lint.json).

Impact: The Field column does not match the path that implementations report.

Suggested fix: Use `choices` in the Field column for both rows.

Grammar violations

Severity	Location	Description
MEDIUM	docs/references/grammar.md:105	**F10** lazy_line accepts feedback and comment lines, so the item grammar accepts two feedback blocks
MEDIUM	docs/question-types/multiple-selection.md:43-44	**F11** No grammar defines the generic bracket that counts as a value for type inference and foreign-choice-marker
LOW	docs/references/grammar.md:103-104	**F9** feedback_line and comment_line accept any indentation before the marker
LOW	docs/question-types/multiple-selection.md:42	**F12** The value bracket is read as a value when a link destination or reference follows it
LOW	docs/question-types/multiple-selection.md:42	**F13** The item rule accepts any indentation before the value

F10: lazy_line accepts feedback and comment lines, so the item grammar accepts two feedback blocks

Location: docs/references/grammar.md:105

```
lazy_line     : ws? md_inline+ nl
```

Description: `md_inline` is any inline element, and a `>` or `!` character is inline text, so `lazy_line` matches a feedback line, a comment line, a list item or a heading. In the multiple-selection `item` rule (`md_inline* nl lazy_line* (blank_line* observations)?`), the lines of a first feedback block can then parse as lazy lines of the choice text. Counter-example, tested with Lark: `[ ] Nile` / `> fb1` / blank line / `> fb2` parses, with `> fb1` as choice text and `> fb2` as the feedback. `[ ] Nile` / `> fb1` / `! c` / `> fb2` also parses. The prose forbids both: a lazy line is a line without a marker (docs/references/grammar.md:127-131) and a second block of the same kind is a `parse-error` (docs/references/grammar.md:134-137, docs/question-types/multiple-selection.md:156). With a stub `lazy_line` that excludes lines starting with `>`, `!` (not `![`), list markers, `#` and fences, the grammar rejects both inputs as the prose requires.

Impact: The grammar does not encode the one-block-of-each-kind rule that it appears to encode; the restriction exists only in prose.

Suggested fix: Make `lazy_line` exclude lines that start a feedback line, a comment line or a block that interrupts a paragraph, for example with a terminal that has a negative lookahead for these starts.

Exploitability conditions: Item with a feedback or comment block.

F11: No grammar defines the generic bracket that counts as a value for type inference and foreign-choice-marker

Location: docs/question-types/multiple-selection.md:43-44

```
value : ws
      | ws? ("x" | "X") ws?
```

Description: The `value` rule rejects `[]`, `[xx]`, `[ x x ]`, `[*]`, `[50%]`, `[v]` and `[U+00A0]` (tested with Lark). docs/question-types/question-base.md:217-222 says a value outside the `value` rule of the type is a `foreign-choice-marker`, and gives `[?]` as an example, and type inference reads "every item starts with `[value]`" before the type is known. No grammar defines that generic `[value]`: it is not stated whether `[xx]`, `[x x]`, `[Brasília]` or `[a b c]` at the start of an item is a foreign value (the list stays a choice list) or text (the list is not a choice list). The Python implementation reports `foreign-choice-marker` for `[xx]` and `[ x x ]`. The `[]` case is a special case of this gap (see F2).

Impact: Implementations can report different codes, or infer a different body, for the same malformed marker.

Suggested fix: Declare a shared rule for the bracket at the start of a choice item (for example `"[" /[^\]\r\n]*/ "]"`, minus what CHOICE_ID or a link would take) that type inference and `foreign-choice-marker` use.

F9: feedback_line and comment_line accept any indentation before the marker

Location: docs/references/grammar.md:103-104

```
feedback_line : ws? ">" ws? md_inline* nl
comment_line  : ws? COMMENT_MARK ws? md_inline* nl
```

Description: The prose (docs/references/grammar.md:117-121) allows at most three spaces of indentation before > or !, relative to the list item content, and says that with four or more the line is not a feedback or comment line. The leading ws? has no bound. Counter-example: the item * [x] Rio followed by the line > fb (four spaces relative to the item content). Tested with Lark: the grammar parses the line as a feedback_line. Under CommonMark, the line is a paragraph continuation, so > fb is part of the choice text. The same applies to ! lines and to a tab before the marker.

Impact: A reader of the grammar alone gets a different choice text and feedback than the prose and CommonMark give. The Python implementation also reads the over-indented line as feedback, which suggests the grammar is what implementers follow.

Suggested fix: Bound the indentation in the rule (for example INDENT: / {0,3}/, with a note on how a tab counts), or state next to the snippet that ws? before a marker is limited by the Marker rule.

Exploitability conditions: Feedback or comment line indented four or more columns relative to the item content.

F12: The value bracket is read as a value when a link destination or reference follows it

Location: docs/question-types/multiple-selection.md:42

```
item      : ws? "[" value "]" ws? (CHOICE_ID ws?)? md_inline* nl lazy_line* (blank_line*
observations)?
```

Description: CHOICE_ID has the lookahead (?![(\[]) because a bracket followed by (or [starts a markdown link (docs/references/grammar.md:83-86). The value bracket has no such lookahead. Counter-examples, tested with Lark: * [x](https://ibge.gov.br) Brasília parses as a correct choice with the text (https://ibge.gov.br) Brasília, and * [x][ref] Brasília parses as a correct choice with the id ref. CommonMark reads [x](https://ibge.gov.br) as an inline link and [x][ref] as a full reference link when ref is defined. The Python implementation reads the first as a choice and the second as an epilogue paragraph (forbidden-block-element).

Impact: Contrived input, but the grammar and the markdown layer disagree on whether the item is a choice.

Suggested fix: State that a value bracket followed by (or [is still the value (and the rest is text), or add the same lookahead to the value bracket and define the result.

F13: The item rule accepts any indentation before the value

Location: docs/question-types/multiple-selection.md:42

```
item      : ws? "[" value "]"
```

Description: docs/references/grammar.md:54-57 says the source text is read at the start of a block and only spaces and tabs are accepted, but sets no limit. Counter-example, tested with Lark: `*  [x]  Rio` (six spaces after the list marker) parses as a correct choice. Under CommonMark, five or more spaces after the marker make the item content an indented code block, so the item has no paragraph that starts with `[x]`. The feedback rules state a three-space limit; the item has none. The Python implementation accepts the input as a choice.

Impact: The grammar accepts a choice that a markdown renderer shows as code.

Suggested fix: State the limit for the `ws?` before the value (CommonMark: at most four spaces after the list marker), or state that the grammar applies to the item content after CommonMark removes the marker and its padding.

Schema mismatches

Severity	Location	Description
LOW	schema/multiple-selection.yaml:33-34	**F23** Schema grading description omits the per-type map of the exam
INFO	schema/multiple-selection.yaml:7	**F24** Schema top-level description lists only `choices` as an extension

F23: Schema grading description omits the per-type map of the exam

Location: schema/multiple-selection.yaml:33-34

Mechanical fix

```
`inherit`, the default, takes the exam's `grading` value, and
          is `symmetric` outside an exam.
```

Description: multiple-selection.md:99-100 and exam.md (Grading) say that `inherit` takes the exam's `grading`, or its `multiple-selection` entry when it is a map, and that a missing key is `symmetric`. The schema description only mentions 'the exam's `grading` value', which is wrong when that value is a map.

Impact: A reader of the schema alone cannot resolve `inherit` against a map-valued exam `grading`.

Suggested fix: Write: '`inherit`, the default, takes the exam's `grading` (its `multiple-selection` entry when it is a map, `symmetric` if the entry is missing), and is `symmetric` outside an exam.'

F24: Schema top-level description lists only `choices` as an extension

Location: schema/multiple-selection.yaml:7

Mechanical fix

```
Extends the common question-base schema with a `choices` field.
```

Description: The schema also adds `grading` and `shuffle` (lines 23-44), which question-base.yaml does not declare, and which the Frontmatter table of multiple-selection.md lists.

Impact: Minor: the description understates the fields of the type.

Suggested fix: Write 'with `choices`, `grading` and `shuffle` fields'.

Prioritized recommendations

P1: Define the leading bracket of a choice item and the type inference fallback, so every malformed marker has one stated result. (Findings: F1, F2, F11, F12)

P2: Tighten the shared item grammar: lazy lines without markers, and a stated indentation bound; add the matching invalid examples. (Findings: F10, F25, F9, F13)

P2: Make the Choices section of multiple selection self-contained, preferably by moving the shared choice rules to one section. (Findings: F3, F4, F14)

P3: Apply the mechanical text and schema fixes in one pass. (Findings: F6, F7, F8, F22, F23, F24)

P3: Clarify the grading table notation, rephrase info rules, and fill the corpus gaps. (Findings: F5, F21, F26, F27)

GitHub issues

Each block below is a complete issue, ready to paste.

````
--- ISSUE 1 ---
# [Spec] Define the bracket at the start of a choice item

**Labels:** medium, low

## Problem

The spec defines each type's `value` rule, but not the generic bracket that type inference and
`foreign-choice-marker` read before the type is known. So `[]`, `[xx]`, a list of only blank
and `[?]` values, and `[x](url)` have no stated result. Options: (a) declare one shared rule
for the leading bracket in grammar.md, relax type inference rule 4 to 'no other rule applies:
multiple selection', and state that `[]` is a `parse-error` (what mdq-py does); (b) treat any
non-matching bracket as text, so the list is not a choice list. Recommended: (a), since it
matches mdq-py and the corpus and never drops a body in silence.

## Evidence

- `docs/question-types/question-base.md:213-216`: Type inference has no rule for a list of
blank and unknown values

  ```md
  1. a value is `*` or a percentage: multiple choice;
  2. a value is `x` or `X`: multiple selection;
  3. a value is a letter: true/false;
  4. every value is blank: multiple selection.
  ```
- `docs/question-types/multiple-selection.md:42-44`: Empty brackets `[]` in a choice list have
no rule

  ```lark
  value : ws
        | ws? ("x" | "X") ws?
  ```
- `docs/question-types/multiple-selection.md:43-44`: No grammar defines the generic bracket
that counts as a value for type inference and foreign-choice-marker

  ```lark
  value : ws
        | ws? ("x" | "X") ws?
  ```
- `docs/question-types/multiple-selection.md:42`: The value bracket is read as a value when a
link destination or reference follows it

  ```lark
  item      : ws? "[" value "]" ws? (CHOICE_ID ws?)? md_inline* nl lazy_line* (blank_line*
observations)?
  ```

## Impact

- **Type inference has no rule for a list of blank and unknown values:** Two implementations
can infer different types, or none, and report different codes for the same document.
- **Empty brackets `[]` in a choice list have no rule:** `[]` is a likely typo for `[ ]`.
Implementations can disagree on the code, and one could read the list as prose and drop the
question body.
- **No grammar defines the generic bracket that counts as a value for type inference and
foreign-choice-marker:** Implementations can report different codes, or infer a different body,
 for the same malformed marker.
- **The value bracket is read as a value when a link destination or reference follows it:**
Contrived input, but the grammar and the markdown layer disagree on whether the item is a
choice.

## Suggested fix

- **Type inference has no rule for a list of blank and unknown values:** Relax rule 4 to 'no
other rule applies: multiple selection' (what mdq-py does), and add an invalid example with
only blank and unknown values.
- **Empty brackets `[]` in a choice list have no rule:** State in question-base.md, Type
inference, that `[]` in a choice list is a `parse-error`, as mdq-py does, and add a multiple-
selection invalid example.
````

```
- **No grammar defines the generic bracket that counts as a value for type inference and
foreign-choice-marker:** Declare a shared rule for the bracket at the start of a choice item
(for example `"[" /[^\]\r\n]*/ "]"`, minus what CHOICE_ID or a link would take) that type
inference and `foreign-choice-marker` use.
- **The value bracket is read as a value when a link destination or reference follows it:**
State that a value bracket followed by `(` or `[` is still the value (and the rest is text), or
 add the same lookahead to the value bracket and define the result.

## Acceptance criteria

- [ ] grammar.md declares the leading bracket rule used by type inference and foreign-choice-
marker
- [ ] question-base.md type inference covers a list of blank and unknown values
- [ ] The spec states the code for `[]` and for a value bracket followed by `(` or `[`
- [ ] Invalid examples cover `[]` and blank-plus-`[?]` in a multiple-selection list

--- END ISSUE 1 ---
```

````
--- ISSUE 2 ---
# [Spec] Make lazy_line exclude marker lines and test the one-block rule

**Labels:** medium

## Problem

`lazy_line` matches `>` and `!` lines, so the item grammar accepts two feedback blocks, which
the prose forbids; no multiple-selection example tests it, and mdq-py merges the blocks. Fix
the terminal with a negative lookahead for marker and block-start lines, and add invalid
examples.

## Evidence

- `docs/references/grammar.md:105`: lazy_line accepts feedback and comment lines, so the item
grammar accepts two feedback blocks

  ```lark
  lazy_line     : ws? md_inline+ nl
  ```
- `docs/question-types/multiple-selection.md:156`: No multiple-selection example has a second
feedback or comment block, and mdq-py merges them

  ```markdown
  | choices[]          | error    | parse-error               | carries at most one feedback
block and one comment block  |
  ```

## Impact

- **lazy_line accepts feedback and comment lines, so the item grammar accepts two feedback
blocks:** The grammar does not encode the one-block-of-each-kind rule that it appears to
encode; the restriction exists only in prose.
- **No multiple-selection example has a second feedback or comment block, and mdq-py merges
them:** The rule is not tested, and the reference implementation violates it without the suite
noticing.

## Suggested fix

- **lazy_line accepts feedback and comment lines, so the item grammar accepts two feedback
blocks:** Make `lazy_line` exclude lines that start a feedback line, a comment line or a block
that interrupts a paragraph, for example with a terminal that has a negative lookahead for
these starts.
- **No multiple-selection example has a second feedback or comment block, and mdq-py merges
them:** Add invalid examples for a choice with two feedback blocks (separated by a blank line,
and by a comment block), expecting `parse-error`.

## Acceptance criteria

- [ ] lazy_line in grammar.md rejects lines that start with `>`, `!` (not `![`), a list marker,
 `#` or a fence
- [ ] Invalid multiple-selection examples with two feedback blocks (separated by a blank line,
and by a comment) expect parse-error
- [ ] spec_audit.py passes

--- END ISSUE 2 ---
````

````
--- ISSUE 3 ---
# [Spec] Bound the indentation in the choice item and feedback grammars

**Labels:** low

## Problem

`ws?` before the value and before `>`/`!` is unbounded, while the prose and CommonMark limit
it. Options: (a) bound it in the rules; (b) state once in grammar.md that item grammars apply
to the item content after CommonMark removes the list marker and its padding, and keep the
three-space rule for markers. Recommended: (b), it keeps the snippets short and matches what
markdown-it gives the implementations.

## Evidence

- `docs/references/grammar.md:103-104`: feedback_line and comment_line accept any indentation
before the marker

  ```lark
  feedback_line : ws? ">" ws? md_inline* nl
  comment_line  : ws? COMMENT_MARK ws? md_inline* nl
  ```
- `docs/question-types/multiple-selection.md:42`: The item rule accepts any indentation before
the value

  ```lark
  item      : ws? "[" value "]"
  ```

## Impact

- **feedback_line and comment_line accept any indentation before the marker:** A reader of the
grammar alone gets a different choice text and feedback than the prose and CommonMark give. The
 Python implementation also reads the over-indented line as feedback, which suggests the
grammar is what implementers follow.
- **The item rule accepts any indentation before the value:** The grammar accepts a choice that
 a markdown renderer shows as code.

## Suggested fix

- **feedback_line and comment_line accept any indentation before the marker:** Bound the
indentation in the rule (for example `INDENT: / {0,3}/`, with a note on how a tab counts), or
state next to the snippet that `ws?` before a marker is limited by the Marker rule.
- **The item rule accepts any indentation before the value:** State the limit for the `ws?`
before the value (CommonMark: at most four spaces after the list marker), or state that the
grammar applies to the item content after CommonMark removes the marker and its padding.

## Acceptance criteria

- [ ] grammar.md states how indentation before the value and the markers is bounded
- [ ] A choice indented as code and a feedback line indented four columns have a stated result

--- END ISSUE 3 ---
````

````
--- ISSUE 4 ---
# [Spec] Make the multiple-selection Choices section self-contained

**Labels:** medium, low

## Problem

Choices imports multiple-choice#choices without saying which parts apply, the info rules on the
 count of correct choices exist only in the table, and the page does not say when a list is
inferred as multiple selection. Options: (a) list the shared parts and add the missing
paragraphs to multiple-selection.md; (b) move the shared choice rules (text, id, feedback,
comment, duplicates, visual equivalence, derived ids) to one shared section that multiple
choice, multiple selection and true/false reference. Recommended: (b), it also removes the risk
 that an edit to multiple choice changes multiple selection.

## Evidence

- `docs/question-types/multiple-selection.md:60-61`: Choices section does not say which parts
of multiple-choice#choices apply

  ```md
  Choices follow the same structure as [Multiple choice questions](multiple-choice.md#choices),
  respecting the rules for the allowed `values` inside the initial square brackets.
  ```
- `docs/question-types/multiple-selection.md:158-159`: Text never states the no-correct-choice
and all-choices-correct lints

  ```md
  | choices[].correct  | info     | no-correct-choice          | at least one choice should be
correct                      |
  | choices[].correct  | info     | all-choices-correct        | not every choice should be
correct                         |
  ```
- `docs/question-types/multiple-selection.md:5-7`: Multiple selection does not state how a
choice list is inferred as multiple selection

  ```markdown
  Multiple selection questions are represented by an unordered list with an ASCII-art
  rendition of a checkbox. The correct answers are marked with `[x]` and incorrect
  ones are left blank.
  ```

## Impact

- **Choices section does not say which parts of multiple-choice#choices apply:** Readers and
porters can apply multiple-choice-only rules (score derivation, single pick, many-correct-
choices) to multiple selection.
- **Text never states the no-correct-choice and all-choices-correct lints:** The rule exists
only in the table; the level and the rationale cannot be checked against the text.
- **Multiple selection does not state how a choice list is inferred as multiple selection:** A
reader of this page cannot tell that a list with no `[x]` is still multiple selection, or that
one `[*]` turns the whole list into multiple choice.

## Suggested fix

- **Choices section does not say which parts of multiple-choice#choices apply:** List the
shared rules explicitly (text, id, feedback, comment, duplicates, visual equivalence, derived
ids) and say that `correct` replaces the `score` row of the AST table; or move the shared
choice rules to one shared section that both types reference.
- **Text never states the no-correct-choice and all-choices-correct lints:** Add a paragraph to
 Choices: any number of choices MAY be correct, including none or all; implementations MAY
report `no-correct-choice` and `all-choices-correct` at the info level, with the reason above.
- **Multiple selection does not state how a choice list is inferred as multiple selection:**
Add a sentence after the example that summarizes the inference rules for multiple selection and
 links to question-base.md#type-inference, as multiple-choice.md does.

## Acceptance criteria

- [ ] multiple-selection.md states which choice rules apply and that score rules do not
- [ ] The text states no-correct-choice and all-choices-correct with MAY at the info level
````

````
- [ ] multiple-selection.md summarizes its type inference and links to question-base.md#type-
inference

--- END ISSUE 4 ---


--- ISSUE 5 ---
# [Spec] Clarify the grading table notation of multiple selection

**Labels:** low

## Problem

The table uses T, F and _ for a two-state question. Recommended: write markings as ticked sets
and drop the 'Expands to' column.

## Evidence

- `docs/question-types/multiple-selection.md:120-132`: The T, F and _ notation of the grading
table is not defined

  ```md
  | `[T, F, _, _]` | `[T, F, F, F]` |        3         |   0.75    |       0.0        |     0.5
     |
  ```

## Impact

Readers can think that multiple selection has a third state, which the same document denies.

## Suggested fix

Either write the markings as ticked sets (`{a, c}`, `{a}`, `{}`) and drop the 'Expands to'
column, or say before the table that `T` and `F` are the `true` and `false` values of the
response mapping and `_` is an absent key (see responses.md#multiple-selection). Recommended:
ticked sets, since both forms are the same response.

## Acceptance criteria

- [ ] The grading table uses a notation the page defines
- [ ] examples/grading/multiple-selection.yaml still agrees with the table

--- END ISSUE 5 ---
````

````
--- ISSUE 6 ---
# [Spec] Phrase info-level rules without 'should'

**Labels:** low

## Problem

The Rule column says 'should' for info rules, which by convention maps to warning. Rephrase
them as observations in multiple-selection.md and multiple-choice.md.

## Evidence

- `docs/question-types/multiple-selection.md:157-160`: Rules table uses 'should' for info-level
 rules

  ```markdown
  | choices[].text     | info     | visually-identical-choices | should not be visually
equivalent to another choice[^2]   |
  | choices[].correct  | info     | no-correct-choice          | at least one choice should be
correct                      |
  | choices[].correct  | info     | all-choices-correct        | not every choice should be
correct                         |
  | choices[].id       | info     | missing-choice-id          | should be defined in the
document, instead of derived[^3] |
  ```

## Impact

A reader that maps the wording to a level gets `warning`, not `info`.

## Suggested fix

Phrase the info rules as observations, for example 'is visually equivalent to another choice',
'no choice is correct', 'every choice is correct', 'has no id of its own'.

## Acceptance criteria

- [ ] No info row of the two tables uses 'should' or 'must'

--- END ISSUE 6 ---
````

````
--- ISSUE 7 ---
# [Spec] Fill the multiple-selection corpus gaps

**Labels:** low

## Problem

Add `.mdq.md` examples for blank text, whitespace-only duplicate texts and multi-line text and
feedback, and grading cases with no or all correct choices and with two choices.

## Evidence

- `docs/question-types/multiple-selection.md:48-50`: Markdown-level blank-text and text-
normalization cases are only covered by YAML examples

  ```markdown
  In
  `* [x] [brasilia]`, `brasilia` is the id and the text is empty, which is a
  `blank-text` error.
  ```
- `examples/grading/multiple-selection.yaml:17-115`: Grading corpus has no question with zero
or all correct choices, or with two choices

  ```yaml
  questions:
    # The table in multiple-selection.md, "Grading": answer key [T, F, T, F].
  ```

## Impact

- **Markdown-level blank-text and text-normalization cases are only covered by YAML examples:**
 The parser path that produces these values for multiple selection is not tested.
- **Grading corpus has no question with zero or all correct choices, or with two choices:**
Edge cases of the grading formulas that the spec allows are not checked against
implementations.

## Suggested fix

- **Markdown-level blank-text and text-normalization cases are only covered by YAML examples:**
 Add `.mdq.md` invalid examples for `* [x] [id]` with no text and for whitespace-only duplicate
 texts, and a valid example with multi-line choice text and feedback.
- **Grading corpus has no question with zero or all correct choices, or with two choices:** Add
 grading cases with keys `{a: false, b: false}` and `{a: true, b: true}` for the three
strategies.

## Acceptance criteria

- [ ] The new examples exist and spec_audit.py passes
- [ ] Grading cases cover {a: false, b: false} and {a: true, b: true} under the three
strategies

--- END ISSUE 7 ---
````

````
--- ISSUE 8 ---
# Mechanical fixes

**Labels:** mechanical, low, info

## Problem

Each fix below follows from its finding and needs no design decision, so they can be applied
together in one pass and reviewed by reading the diff. Severity still applies: a mechanical fix
 can matter a lot.

### F6 (low): 'no way to explicitly leave it unmarked' reads as the opposite of its intent

`docs/question-types/multiple-selection.md:137-141`

```md
There
is no way to explicitly leave it unmarked. If that is necessary, prefer using
[true/false](true-false.md) questions.
```

**Suggested fix:** Reword: 'A student cannot abstain on a single choice: an unticked choice is
a false verdict. If abstention is needed, use true/false questions.'

### F7 (low): 'When neither ... and outside an exam' joins two cases with 'and'

`docs/question-types/multiple-selection.md:100-102`

```md
When neither
the question nor the exam sets a strategy, and outside an exam, the strategy
is `"symmetric"`.
```

**Suggested fix:** Write: 'The strategy is `"symmetric"` outside an exam, and in an exam that
does not set one for multiple selection (no `grading`, or a map without a `multiple-selection`
key).' Apply the same wording in multiple-choice.md.

### F8 (low): Body does not state the two-choice minimum, unlike multiple choice

`docs/question-types/multiple-selection.md:39`

```md
Body consists of an unordered list of items. Each item follows the grammar:
```

**Suggested fix:** Write 'an unordered list of at least two items'.

### F22 (low): Rules table field for count rules is a choice field, but the code reports on the
 list

`docs/question-types/multiple-selection.md:158-159`

```markdown
| choices[].correct  | info     | no-correct-choice          | at least one choice should be
correct                     |
```

**Suggested fix:** Use `choices` in the Field column for both rows.

### F23 (low): Schema grading description omits the per-type map of the exam

`schema/multiple-selection.yaml:33-34`

```yaml
`inherit`, the default, takes the exam's `grading` value, and
          is `symmetric` outside an exam.
```

**Suggested fix:** Write: ``inherit`, the default, takes the exam's `grading` (its `multiple-
````

````
selection` entry when it is a map, `symmetric` if the entry is missing), and is `symmetric`
outside an exam.'

### F24 (info): Schema top-level description lists only `choices` as an extension

`schema/multiple-selection.yaml:7`

```yaml
Extends the common question-base schema with a `choices` field.
```

**Suggested fix:** Write 'with `choices`, `grading` and `shuffle` fields'.

## Acceptance criteria

- [ ] F6: 'no way to explicitly leave it unmarked' reads as the opposite of its intent
- [ ] F7: 'When neither ... and outside an exam' joins two cases with 'and'
- [ ] F8: Body does not state the two-choice minimum, unlike multiple choice
- [ ] F22: Rules table field for count rules is a choice field, but the code reports on the
list
- [ ] F23: Schema grading description omits the per-type map of the exam
- [ ] F24: Schema top-level description lists only `choices` as an extension

--- END ISSUE 8 ---
````